matlab.mathworks.com

HOME PLOTS APPS EDITOR PUBLISH VIEW

Search (Ctrl+Shift+Space) Janani

New Open Save Compare Go To Find Bookmark Copilot Refactor Code Issues Debugger Generate Test Find Tests Run Section Section Break Run and Advance Run to End Run Step Stop

FILE NAVIGATE CODE ANALYZE TEST SECTION RUN

/ > MATLAB Drive >

Editor

untitled8.m × untitled10.m * ×

/MATLAB Drive/untitled10.m

```
clear;
clc;
close all;

freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;

bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];

txPowerdBm = 30;
gtDbi = 0;
grDbi = 0;

distanceKm = logspace(log10(0.1), log10(2), 200);
distanceAtPointKm = 0.5;

fspl = @(fMHz,dKm) 32.44 + ...
    20*log10(fMHz) + 20*log10(dKm);

lossAtPoint = fspl(freqMHz, distanceAtPointKm);

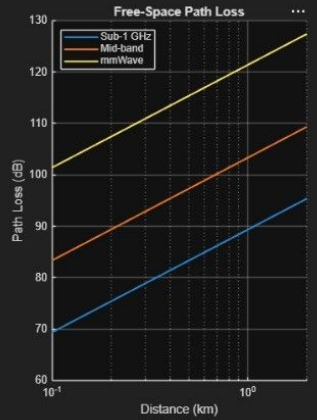
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;

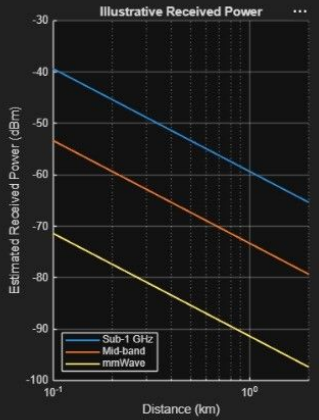
linkTable = table( ...
    bandName, freqGHz(:), lossAtPoint(:), rxAtPoint(:), ...
    'VariableNames', ...
    {'Band', 'Frequency_GHz', 'FSPL_dB', 'EstimatedRx_dBm'});

disp('Link Budget Results:');
```

Figures

Figure 1 × Figure 2 ×





Command Window

New to MATLAB? See resources for Getting Started.

```
    Band          Rx_dBm
    ____________    ______

    "Sub-1 GHz"     -53.321
    "Mid-band"      -67.301
    "mmWave"        -85.363

>>
```